Exp No: 4	*Setup and configure a LAN using a Switch and Ethernet cables*

AIM:

To set up and configure a LAN using a switch and Ethernet cables, assign IP addresses to the hosts, check the connectivity using the ping command and share files and folders across the machines of the LAN.

ALGORITHM / PROCEDURE:

1. Connect 4 host machines to the switch using straight-through UTP cables (RJ45). Check that the link LEDs of the switch glow.
2. Assign a static IP address to each host: ***Windows***, Control Panel > Network and Sharing Centre > Change adapter settings > Ethernet > Properties > IPv4 (or run ncpa.cpl).
3. Use the addresses 192.168.10.11 to 192.168.10.14 with subnet mask 255.255.255.0. Leave the default gateway empty because there is only one network.
4. Open the Command Prompt and verify the settings with ipconfig.
5. Check the connectivity between the machines with ping, for example from PC1 ping `192.168.10.12`.
6. On PC1 create a folder LabShare, right click > Properties > Sharing > Advanced Sharing > tick ***Share this folder*** and give ***Read/Write*** permission to ***Everyone***.
7. On PC2 open ***File Explorer*** and type \\192.168.10.11 in the address bar. The files of PC1 are displayed.
8. The folder can also be mapped as a drive with net use Z: \\192.168.10.11\LabShare and the shared files can be copied between the machines.
9. Note down the output of each command.

Host	*IP address*	*Subnet mask*	*Switch port*
PC1	192.168.10.11	255.255.255.0	Fa0/1
PC2			Fa0/2
PC3	192.168.10.12	255.255.255.0	Fa0/3
PC4	192.168.10.13	255.255.255.0	Fa0/4
	192.168.10.14	255.255.255.0	

DIAGRAM:

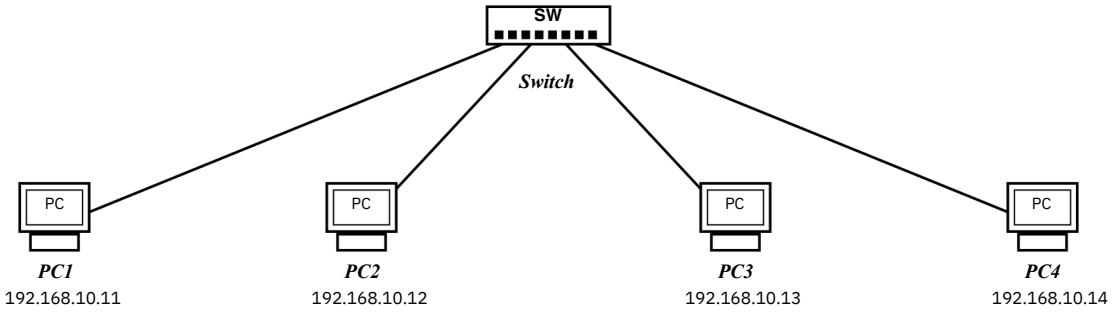


Fig 1: LAN with 4 hosts connected to a switch (star topology)

OUTPUT:

Command Prompt - PC1

```
C:\Users\UusSeErR>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix . :
   Link-local IPv6 Address . . . . . : fe80::6c1f:3a2e:91b4:d7c2%8
   IPv4 Address. . . . . . . . . . . : 192.168.10.11
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

Command Prompt - PC1

```
C:\Users\UusSeErR>ping 192.168.10.12

Pinging 192.168.10.12 with 32 bytes of data:
Reply from 192.168.10.12: bytes=32 time=1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\UusSeErR>ping 192.168.10.13

Pinging 192.168.10.13 with 32 bytes of data:
Reply from 192.168.10.13: bytes=32 time<1ms TTL=128
Reply from 192.168.10.13: bytes=32 time<1ms TTL=128
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Reply from 192.168.10.13: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\UusSeErR>ping 192.168.10.14

Pinging 192.168.10.14 with 32 bytes of data:
Reply from 192.168.10.14: bytes=32 time<1ms TTL=128
Reply from 192.168.10.14: bytes=32 time<1ms TTL=128
Reply from 192.168.10.14: bytes=32 time<1ms TTL=128
Reply from 192.168.10.14: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

Command Prompt - PC2

```
C:\Users\UusSeErR >net view \\192.168.10.11
Shared resources at \\192.168.10.11

Share name   Type   Used as   Comment
-------------------------------------------------
LabShare     Disk
The command completed successfully.

C:\Users\UusSeErR >net use Z: \\192.168.10.11\LabShare
The command completed successfully.

C:\Users\UusSeErR >dir Z:
 Volume in drive Z is LabShare
Directory of Z:\

07-10-2026  08:42 PM    <DIR>          .
```

Command Prompt - PC2 (contd.)

```
07-10-2026  08:42PM     <DIR>          ..
07-10-2026  08:40PM             14,336 CN_Lab_Notes.docx
07-10-2026  08:41PM            212,480 Exp4_Topology.png
07-10-2026  08:42PM              1,024 hosts_list.txt
               3File(s)         227,840 bytes
               2Dir(s) 48,213,327,872 bytes free

C:\Users\UusSeErR >copy Z:\hosts_list.txt C:\Users\UusSeErR\Desktop
        1 file(s) copied.
```

RESULT:

Thus a LAN was set up with 4 hosts connected to a switch using Ethernet cables, the IP addresses were assigned, the connectivity was verified using the ping command and files were shared and accessed across the machines of the LAN.